

LECTURE 2.1



# Fitting a Line

ECONXXXXX Econometrics I

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# Where We Are

## 1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

## 2. Simple Regression

**2.1 Fitting a Line**

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

## 3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

## 4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

### Block 2 Is Four Lectures on One Estimator

Lectures already given are greyed and today's is marked. Today fits the line. Lecture 2.2 asks how much of earnings it accounts for, 2.3 asks how far it lands from the truth, and 2.4 asks how to report it.

## What Lectures 1.2 and 1.3 Left Open

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Lecture 1.2 said the best guess of earnings given schooling is a **conditional expectation**,  $E[y | x]$ , and that it can be any shape at all.

Lecture 1.3 said any number computed from 120 people is one draw from a distribution, and will miss the population number it estimates.

Today puts those together: we choose one shape, fit it to 120 people, and get a number that misses.

### Question

Given one person's years of schooling and nothing else, what would you guess they earn, and what would you need to know to make that guess?

# Lecture Plan

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## 1 The Line and What It Is For.

A line is a claim about the average earnings of people with a given schooling.

## 2 Least Squares.

Out of every line we could draw, why this one, and where its formula comes from.

## 3 Reading the Slope.

What the number means, in the units the data came in.

## SECTION 1

# The Line and What It Is For

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1. The Line and What It Is For
2. Least Squares
3. Reading the Slope

## A Line Is the Simplest Conditional Expectation

Lecture 1.2's conditional expectation function could bend, jump, or flatten out at any level of schooling. A line refuses all of that. It commits to one claim:

*one more year of schooling changes expected earnings by the same amount, whether it is the ninth year or the nineteenth.*

That is a restriction, not a discovery. We impose it because a line has two numbers, and two numbers can be estimated from 120 people and reported in a sentence. Angrist & Pischke (2015) build the regression chapters of *Mastering Metrics* on exactly this move.

### The Question the Line Answers

Given a person's years of schooling and nothing else, what is our best guess of their hourly earnings?

# The Simple Regression Model and Its Notation

Before choosing a line we write down what one claims: one person, one line, and everything about that person the line does not know.

$$y_i = \beta_0 + \beta_1(x_i - 13) + u_i \quad (1)$$

- $y_i$ : hourly earnings of person  $i$ , in euro
- $x_i$ : years of schooling of person  $i$
- $\beta_0$ : expected earnings at thirteen years of schooling
- $\beta_1$ : the change in expected earnings for one more year
- $u_i$ : everything else that moves person  $i$ 's earnings

## Thirteen Years Is the Leaving Certificate

Eight years of primary, three of junior cycle and two of senior cycle is thirteen. Subtracting it makes  $\beta_0$  the expected earnings of someone who finished school there. Without it,  $\beta_0$  would be expected earnings at zero years of schooling, which describes nobody.

## When the Line Is the Conditional Expectation

The line and the CEF are the same object under one condition, and the rest of the module is an argument about whether it holds (Wooldridge, 2020):

$$E[u | x] = 0 \implies E[y | x] = \beta_0 + \beta_1(x - 13) \quad (2)$$

- $E[u | x] = 0$ : at every level of schooling, everything else has average zero

Take the expectation of the model at a fixed  $x$ . The two  $\beta$  terms are constants, so they pass through unchanged, and  $u$  contributes its conditional average. If that average is zero, what is left is the line.

### Everything Else Must Not Differ With Schooling

If people who stayed longer in school also have higher ability, then  $E[u | x]$  rises with  $x$  and is not zero. The line is then not the CEF, and the slope is not what it appears to be. Lecture 3.1 puts ability into the model; Lecture 4.2 is about when that cannot be done.

# The Hat on the Slope Marks a Sample Estimate

Lecture 1.3 put a hat on an estimate of  $E[y]$ . The same mark now goes on an estimate of a slope, and it means exactly what it meant then.

- $\beta_1$ : one fixed number in the population of 200,000, which we never see
- $\hat{\beta}_1$ : what we compute from our 120 people, which changes with every new sample

## A Sentence With a Hat Is About Your Data

A sentence containing  $\beta_1$  is a claim about the world. A sentence containing  $\hat{\beta}_1$  is a claim about the 120 rows in front of you. The whole of Lecture 2.3 is about the distance between the two.

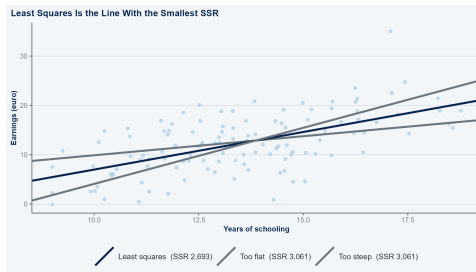
## SECTION 2

# Least Squares

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1. The Line and What It Is For
2. Least Squares
3. Reading the Slope

# Comparing Candidate Lines Through the Same Scatter



**Figure 1:** Three candidate lines through the same 120 people.

## Three Lines, Three Totals

Every line leaves each person a gap. The total of the squared gaps is printed beside each line and the three totals differ, so the totals rank the lines. Nothing so far says that ranking by this total is the right thing to do.

# The Least-Squares Criterion

For any candidate line, every person has a **residual**: the gap between what they earn and what the line says they earn (Wooldridge, 2020).

$$SSR(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1(x_i - 13))^2 \quad (3)$$

- $b_0, b_1$ : any candidate pair, not yet the chosen ones
- $r_i = y_i - b_0 - b_1(x_i - 13)$ : the residual left for person  $i$
- SSR: the total of the squared residuals, over all 120 people

## Least Squares Is a Chosen Criterion

Least squares is the pair  $(b_0, b_1)$  that makes this total as small as possible. That is the entire definition, and it is a criterion someone chose rather than a law.

# Why the Residuals Are Squared

Squaring is a choice with three reasons behind it, and one cost.

- **Raw gaps cancel.** Positive and negative residuals sum to zero for any line through the sample means, so the plain total measures nothing.
- **Absolute values do not cancel, but resist calculus.**  $|r|$  has no derivative at zero, so the minimum has to be searched for numerically.
- **Squares are differentiable everywhere.** Two derivatives, set to zero, give two linear equations, and those solve by hand for a formula.

## The Cost: One Person Can Dominate the Total

A residual of 10 adds 100; ten residuals of 1 add 10 between them. In our 120, the largest residual is 17.23, contributing 296.9 of the total 2,692.8. One person supplies eleven per cent of the criterion.

## Deriving the Slope: Two Conditions

Setting the two derivatives of the SSR to zero gives two conditions on the residuals of the chosen line (Wooldridge, 2020):

$$\sum_{i=1}^n r_i = 0 \qquad \sum_{i=1}^n (x_i - 13) r_i = 0 \qquad (4)$$

The first says the residuals average to zero. The second says they carry no remaining co-movement with schooling: whatever schooling could explain, the line has already used.

### These Two Conditions Are the Whole Derivation

Everything below follows from them by summation algebra alone. No expectations, no calculus beyond the two derivatives already taken.

## Deriving the Slope: Solving the Pair

Expand the first condition. The sum of  $y_i$  is  $n\bar{y}$ , and the sum of a constant is  $n$  times it, so dividing by  $n$  gives the intercept in terms of the slope:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1(\bar{x} - 13) \quad (5)$$

Substitute that into the second condition and the constants cancel in pairs, leaving sums of deviations from the means:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\widehat{\text{Cov}}(x, y)}{\widehat{\text{Var}}(x)} \quad (6)$$

### The Slope Is a Sample Covariance Divided by a Sample Variance

Both were computed on this same sample in Lecture 1.1. Dividing top and bottom by  $n - 1$  turns the sums into the covariance and the variance, which is why the centring constant disappears: subtracting thirteen shifts  $x_i$  and  $\bar{x}$  by the same amount.

## The Slope on Our 120 People

The two sums, computed on the sample in front of us:

$$\hat{\beta}_1 = \frac{965.3}{633.9} = 1.52 \qquad \hat{\beta}_0 = 12.25 - 1.52(13.45 - 13) = 11.57 \qquad (7)$$

- 965.3: the sum of cross-products of deviations,  $\sum(x_i - \bar{x})(y_i - \bar{y})$
- 633.9: the sum of squared deviations of schooling,  $\sum(x_i - \bar{x})^2$

### The Denominator Must Not Be Zero

If every person in the sample had the same schooling,  $\sum(x_i - \bar{x})^2$  would be zero and there would be no slope to compute. A line needs schooling to vary, and how much it varies is what Lecture 2.3 turns into precision.

# Residuals as Vertical Distances From the Fitted Line



**Figure 2:** The least-squares line and the 120 residuals it leaves.

## Every Person Has One Residual

Each segment runs vertically from a person to the line, because the criterion is about  $y$  and not about distance in the plane. On our 120 the squared segments total 2,692.8, and no other line makes that total smaller.

## Squares Rather Than Absolute Values

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Minimising the total absolute gap also picks a line, and it does not pick the same one. It is a real estimator, least absolute deviations, used when a few extreme observations should not be allowed to dominate.

### Question

Which of the two would you rather fit to earnings data, and what would you need to know about the extreme earners before you could answer?

## SECTION 3

# Reading the Slope

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1. The Line and What It Is For
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## The Slope Is Euro Per Year

The slope is a ratio of two units: euro per hour on top, years of schooling underneath. So  $\hat{\beta}_1$  is measured in euro per hour, per year of schooling (Wooldridge, 2020).

### Reading the Estimate Out Loud

Comparing two people in this sample whose schooling differs by one year, the one with more schooling earns about **1.52 euro more per hour**, on average. A person in this sample with thirteen years of schooling earns about **11.57 an hour**.

## The Estimate Missed, and Nothing Went Wrong

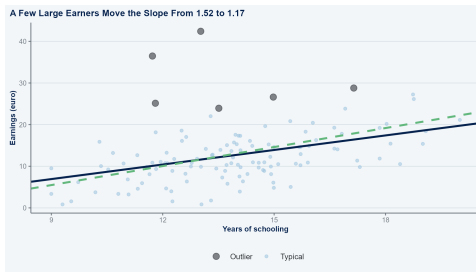
The population slope is 1.30 and we estimated 1.52. The population intercept is 10.80 and we estimated 11.57.

Both misses come from the same source: our 120 people are one draw, and a different 120 would give different numbers. That is Lecture 1.3's sampling variation arriving on a real estimate. Lecture 2.3 puts a number on how much  $\hat{\beta}_1$  varies across samples, and 2.4 reports it alongside the estimate.

### A Fitted Line Is Not an Effect

That sentence does not say a year more school would raise a person's pay by 1.52 euro. Simple regression is an analysis of correlation between two variables, so care is needed in inferring causality (Wooldridge, 2020). Lecture 4.2 is about why.

# How a Few High Earners Change the Fitted Slope



**Figure 3:** The same 120 people, with three very high earners added by hand.

## Squaring Is Why Three People Matter

Solid is the line without the three; dashed is the line with them. The slope rises: a large residual contributes its square, so tilting the line towards those three reduces the total more than it raises it elsewhere.

## Dropping the Highest Earners

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The slope rose because three people sit far above the line. One response is to remove them and refit.

### **Question**

Should those people be dropped from the sample, and what would you need to know about them before you could answer?

# What Would Have To Be True?

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This time the question is asked of the slope we just read out loud, 1.52 euro per hour per year of schooling.

## **Question**

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

## Application

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The app draws the scatter and lets you move a line by hand, and shows the SSR updating as you move it, before the least-squares line appears.

### Question

If we drop the three highest earners, does the fitted slope go up, go down, or stay where it is?

The lab for this lecture is the first regression in R: `lm()`, coefficients and fitted values.

# Summary: The Line, Least Squares, and the Slope

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## 1 The Line and What It Is For.

A line is the simplest conditional expectation there is, and it is the CEF only when  $E[u | x] = 0$ .

## 2 Least Squares.

Squares are chosen because raw gaps cancel and absolute values have no closed form. The price is that one person can dominate the total.

## 3 Reading the Slope.

Covariance over variance gave 1.52 euro per hour per year, against a population 1.30.

## Next Lecture

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Lecture 2.2 asks how much of the variation in earnings this line accounts for:

- **Goodness of Fit:** what  $R^2$  counts, and what it does not.
- **The Hold-Out Sample:** the second 120, opened for the first time.

A line can account for very little of the variation and still carry the slope we want, so a low  $R^2$  is not on its own a verdict.

## Before Lecture 2.2

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### 1 Play:

The app. Fit a line by eye, watch the SSR, then reveal the least-squares one.

### 2 Apply:

Problem set 2, covering 2.1 and 2.2. It repeats today's derivation with  $n = 6$  and numbers you can do by hand.

### 3 Review:

Wooldridge (2020), sections 2-1 to 2-3, on the model, the estimates and their algebra.

# Thank you

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BACKUP

# Appendix

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## Derivation: The Two Derivatives

The body frames start from the two conditions. This is where they come from. Differentiate the SSR with respect to each unknown in turn:

$$\frac{\partial \text{SSR}}{\partial b_0} = -2 \sum_{i=1}^n r_i \qquad \frac{\partial \text{SSR}}{\partial b_1} = -2 \sum_{i=1}^n (x_i - 13) r_i \qquad (8)$$

### Setting Both to Zero Gives the Two Conditions

The  $-2$  divides out, leaving the two sums used on the body frames. The SSR is a sum of squares in  $b_0$  and  $b_1$ , so this stationary point is a minimum rather than a maximum.

## Notes and Tools

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- **The App:** the line-fitting app, and the script that draws this deck's exhibits.
- **The Lecture in One Line:** `coef(lm(y ~ I(x - 13)))`.
- **This Lecture's Lab:** `lm()`, `coef()` and `fitted()`, and nothing else yet.

# References

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Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.